Temi Otun

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EDUCATION

University of Alberta **Expected April 2027**
Bachelor of Science, Major in Computing Science –Artificial Intelligence Specialization, Minor in Business *Edmonton, AB*

- **Relevant Coursework:** Algorithms I, Machine Learning II, Software Engineering, Database Management, Reinforcement Learning, Linear Algebra II, Calculus III, Artificial Intelligence, Applied Statistics II, Computer Architecture I, Ethics of Data Science and AI, Search and Planning in AI
- **Awards**: Jason Lang Scholarship, Dean's List (2025-2026)

EXPERIENCE

Data Engineering Intern **May 2026 – August 2026**
RBC Amplify *Vancouver BC*

- Will collaborate with a cross-functional student team to design and prototype a data-driven solution addressing a real RBC business challenge through rapid iteration and experimentation
- Expected to contribute to data engineering workflows, including pipeline development, data modeling, and cloud-based ETL processes to support analytics and product integration for the final Amplify project demo

Machine Learning Intern **January 2025 – August 2025**
The Metabolomics Innovation Centre *Edmonton AB*

- Enhanced long-range temperature forecasting system, resulting in up to **47%** lower MAE with an average **24.3%** improvement across a six-year test set compared to previous approaches
- Developed multivariate time series models for precipitation forecasting, incorporating **100+** climate features and outperforming baseline models in **73%** of test years
- Assembled forecasting pipelines with advanced feature engineering and benchmarked **20+** ML models, attaining a **42%** accuracy improvement over baseline methods
- Automated SQL pipelines to ingest weather and climate data from APIs and web scrapers, generating new feature combinations to support testing and improvement of forecasting models

Machine Learning Research Assistant **September 2024 – April 2026**
University of Alberta *Edmonton AB*

- Contributed to **3** machine learning systems in computational psychiatry and predictive healthcare, including dementia detection and ECG signal modeling on large scale clinical datasets, improving diagnostic accuracy
- Synthesized insights from **40+** research seminars, on survival analysis and disease prediction, applying advanced statistical and ML methods to strengthen ongoing projects
- Documenting and analyzing **15+** ML experiments, applying feature engineering, hyperparameter tuning, and evaluation pipelines to improve performance across classification and regression tasks

Data Management Intern **January 2024 – May 2024**
InfoStrux *Vancouver, BC*

- Optimized **25+** SQL queries in Snowflake, reducing execution time by up to **60%** and boosting performance of business intelligence dashboards
- Partnered with a senior data engineer to design Snowflake staging and curated layers for 3 datasets; wrote and tuned **30+** queries, improving data reusability and cutting time-to-insight by **40%**
- Constructed and maintained **10** database schemas in Snowflake to support diverse data types, improving pipeline efficiency and data flow

PROJECTS

Emotion Detection CNN | *PyTorch, Computer Vision, Deep Learning, OpenCV, AI, Image Classification, Data Processing, Python* | *Github*

- Trained a deep convolutional neural network (CNN) in PyTorch on the FER 2013 dataset **(32k+** labeled images), utilizing OpenCV for multi-class facial emotion recognition
- Achieved **70%** test accuracy across **7** emotion classes, outperforming baseline models by **15%**, and integrated webcam inference to enable real-time emotion detection
- Incorporated preprocessing techniques (grayscale normalization, resizing, augmentation) with RELU activations and batch normalization, to improve model accuracy

Personal Website | *React, Tailwind CSS, JavaScript, Frontend & Backend Development, Full-Stack, Vercel, Render, REST API Integration, AI* | *Github*

- Launched a personal portfolio with React and Tailwind CSS, hosted on Vercel with backend email integration on Render, providing a platform showcasing AI/ML and software projects
- Improved user experience with interactive UI features, leveraging Framer Motion, Vanta.js, and React-Scroll for animations and dynamic backgrounds

*L**ung Cancer Detection*** | *Python, Machine Learning, Classification, Scikit-learn, AI, SMOTE, Cross Validation, AI engineering* | *Github*

- Accomplished a recall score of **99%**, accuracy score of **94%**, precision score of **95%**, and f1 score of **97%** on the best classification model
- Built and compared multiple ML models (SVM, k-NN, Random Forest) to determine the most effective classification approach
- Applied preprocessing techniques including SMOTE, and k-fold cross-validation to address class imbalance and improve model performance

Research | *Machine Learning & Data Engineering, Python, Audio Data Processing, Feature Engineering, Classification, Model Evaluation, AI*

- Refined ML models on audio datasets for early detection of dementia and mild cognitive impairment, tackling both classification and regression tasks
- Developed a Random Forest model for the ICASSP 2025 SPGC challenge, achieving accurate patient classification into **3** diagnostic categories with evaluation scores
- Explored self-supervised and pre-trained models from prior research, improving predictive metrics (F1, recall, precision, and RMSE) compared to baseline models

SKILLS

Programming: C, Java, Python, SQL
Libraries & Frameworks: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, OpenCV, Matplotlib, Darts, Nixtla, React, Tailwind CSS
Tools & Platforms: Git, Linux, Bitbucket, Docker, Snowflake
Spoken Languages: English, Yoruba